DVLSI Interview questions

Section 1: Digital Design

Q1. MUX-Based Design

- Design XOR using 2:1 MUX
- Minimum MUX required
- Implement XOR using NAND only
- Compare delay

Q2. Flip-Flop Design

- Design mod-6 counter using DFF
- Handle invalid states
- Effect of no reset

Q3. Hazards

- Identify hazard in: F = AB + A'C
- Remove hazard
- Real hardware impact

Q4. Metastability

- Define metastability
- 2-flop synchronizer
- Limitations

Section 2: Verilog HDL

Q5. Coding

- Write 4:1 MUX (behavioral)
- Convert to dataflow
- Hardware inferred

Q6. Debugging

```
always @(a)
   y = a & b;
```

- Identify issue
- Simulation vs synthesis

Q7. Latch Issue

```
always @(a or b)
begin
   if (a)
      y = b;
end
```

- Hardware inferred
- Fix code
- Why latches are risky

Q8. FSM

- Design sequence detector (1011)
- Mealy vs Moore

Section 3: Programming

Q15. Swap

- Without third variable
- XOR method
- Overflow issue

Q16. Arrays

- First non-repeating element
- Optimize approach

Q17. Bit Logic

- Power of 2 check
- Bitwise explanation
